LI

1.10.2025

Porti logice

De luat:

NOT: $y : \Sigma_2 \to \Sigma_2$

A	Y
0	
1	

$\Sigma_2 = \{0, 1\}$

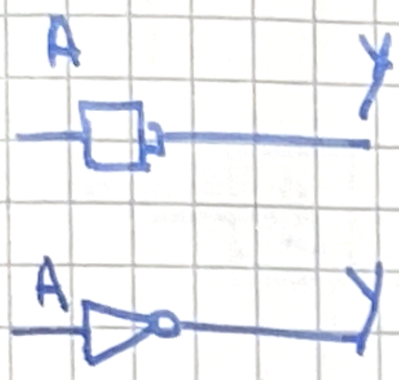
$\Sigma_4 = \{0, 1, 2, 3\}$

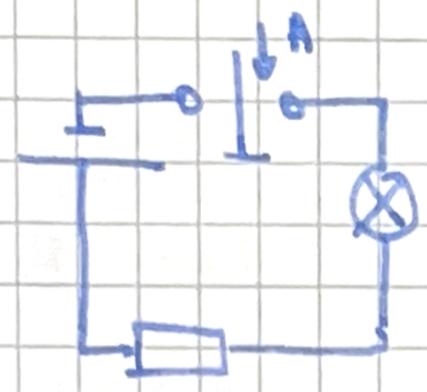
$\Sigma_8 = \{0, 1, 2, 3, 4, 5, 6, 7\}$

$\Sigma_{10} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$\Sigma_{16} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F\}$

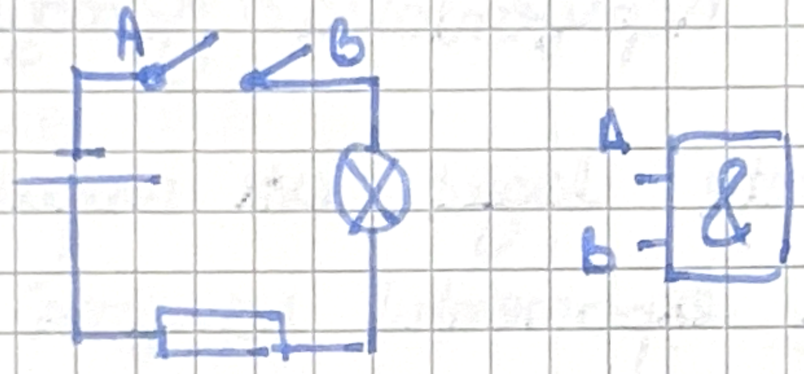
$\Sigma_m = \{0, \overline{(m-1)}\} \in \mathbb{Z}^*$



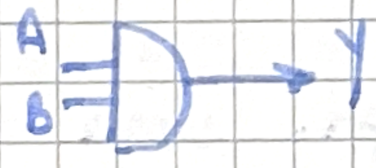


AND $y = A \cdot B$

A	B	y
0	0	0
0	1	0
1	0	0
1	1	1

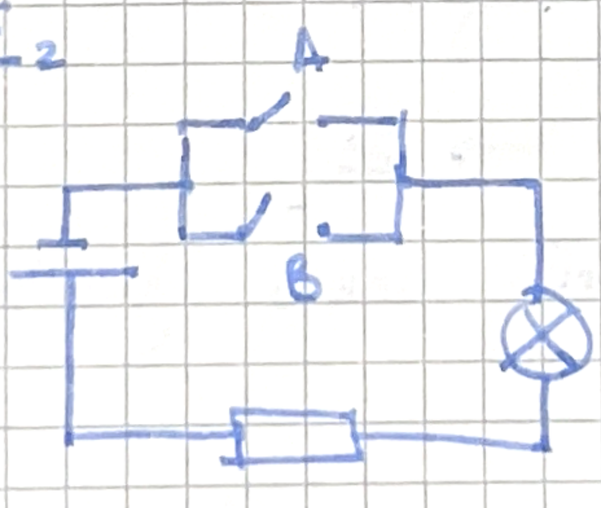


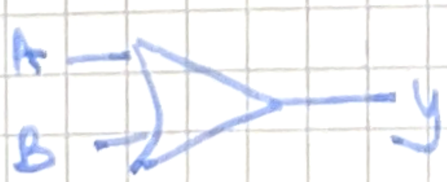
$y : \Sigma_2^2 \to \Sigma_2$

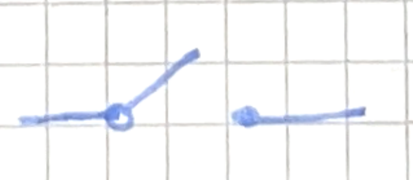


OR $y : \Sigma_2^2 \to \Sigma_2$

A	B	y
0	0	0
0	1	1
1	0	1
0	1	1
1	1	1





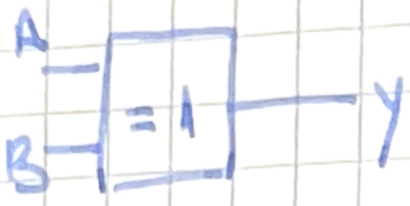


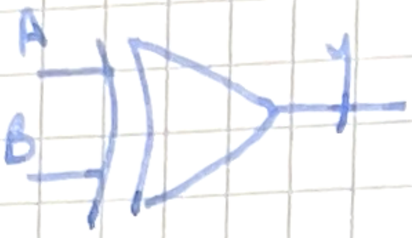
Normal open

Normal close

XOR : $y : \Sigma_2^2 \to \Sigma_2$

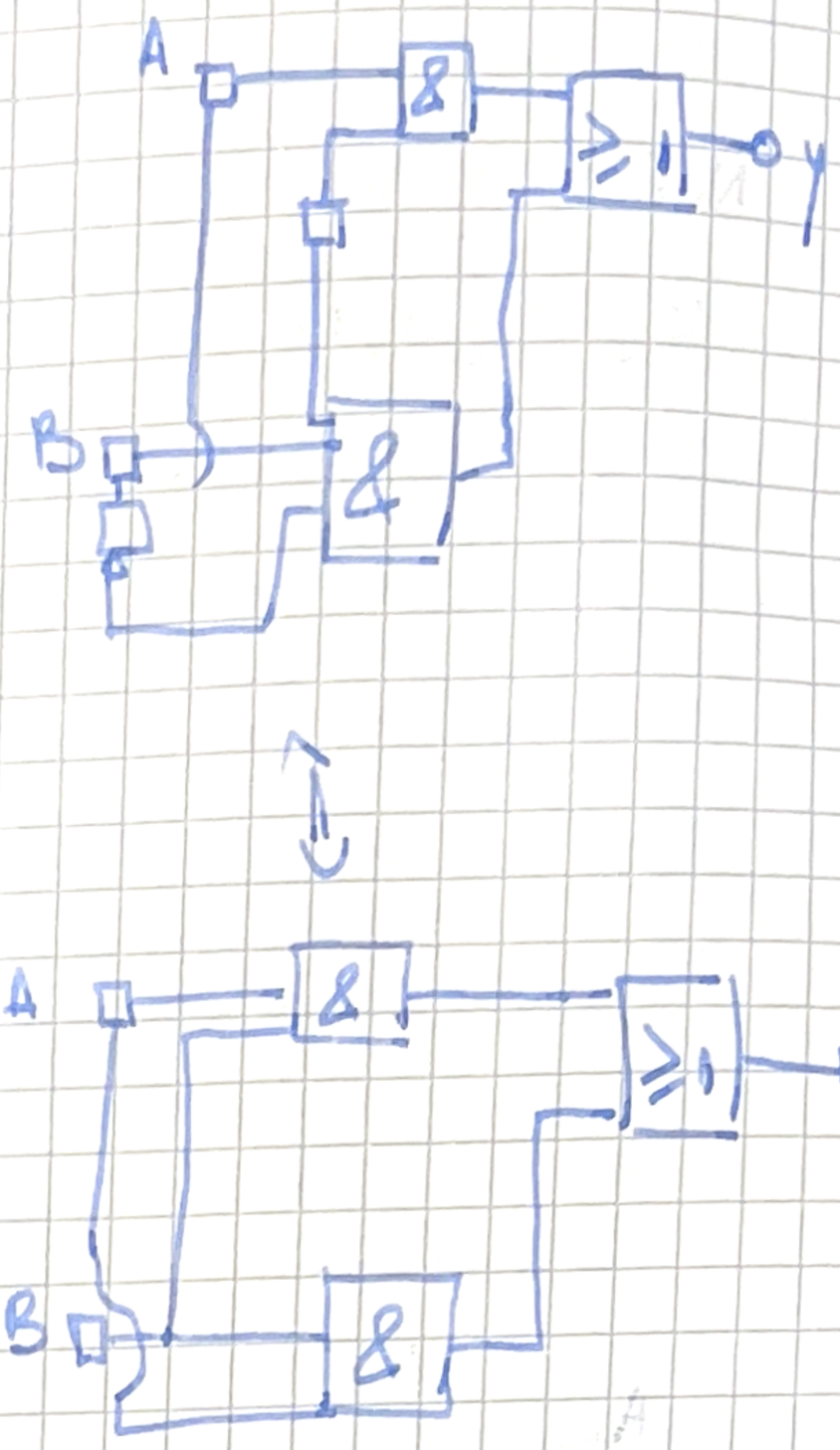
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0





$y = \bar{A}B + A\bar{B}$

$y = \bar{A}B + A\bar{B}$



L2

Reprezentarea funcțiilor logice

O funcție logică este mapată peste inelul nr. binare ($\Sigma_2^{nr.vab}$) iar exponentul reprezintă nr. de variabile

Pentru repr. unei funcții logice putem folosi
- Forma algebrică

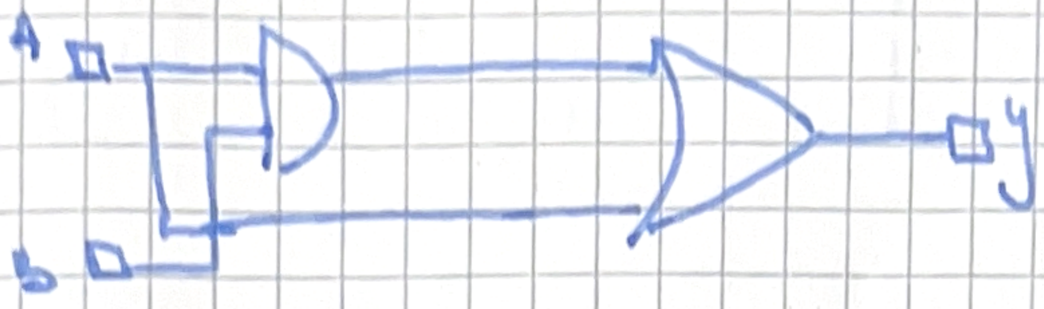
$y : \Sigma_2^2 \to \Sigma_2$

$y : A + AB$ are o formă restrânsă

- Tabela de adevăr : Ea acoperă întreg spațiul de valori de intrare (b^m) și asociază valoarea de ieșire

A	B	y
0	0	0
0	1	0
1	0	1
1	1	1

- Diagramă



https://lg.ccpr.ro

Teoreme

$A + 1 = 1$ ← Sau

$A \cdot \emptyset = \emptyset$

$A \cdot 1 = A$

$\overline{AB} = \overline{A} + \overline{B}$

$\overline{A+B} = \overline{A} \cdot \overline{B}$

+ SAU

$\emptyset$ Și

=== aceeași locație în memorie

același pointer

Asociativitate

$$\overbrace{A+B+C}^{x} = \overbrace{(A+B)}^{y}+C = A+\overbrace{(B+C)}^{z}$$

$x, y, z : \Sigma_2^3 \to \Sigma_2$

$x = A+B+C$

$y = A+(B+C)$

$y = (A+B)+C$

Tabela de adevăr

A	B	C	A + B	b + c	x	y	z
0	0	0	0	0	0	0	0
0	0	1	0	1	1	1	1
0	1	0	1	1	1	1	0
0	1	1	1	0	1	1	1
1	0	0	1	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	1	1	1	1	1
1	1	1	1	1	1	1	0

$\overline{A+B} = \overline{A} + \overline{B}$

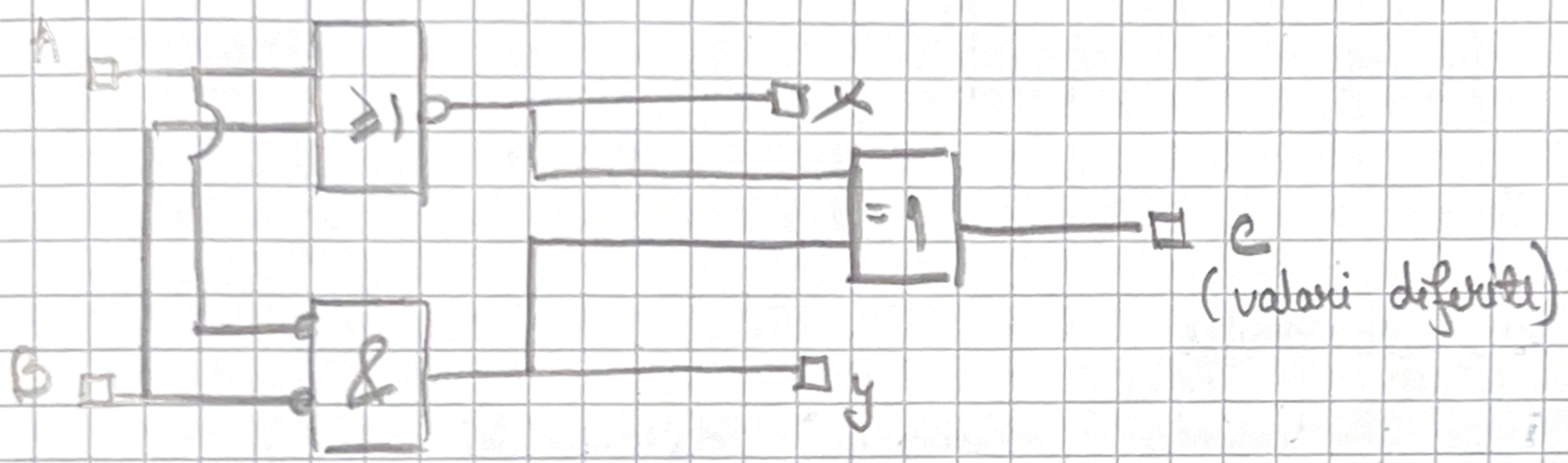
$x, y : \Sigma_2 \xrightarrow{\text{scot}} \Sigma_2$ — Funcții logice cu 2 variabile

$\overline{A}$ = „A negat"

$x = \overline{A+B}$

$y = \overline{A}\,\overline{B}$

A	b	A+B	$\overline{A}$	$\overline{B}$	$\overline{A}\cdot\overline{b}$	$\overline{A+B}$
0	0	0	1	1	1	1
0	1	1	1	0	0	0
1	0	1	0	1	0	0
1	1	1	0	0	0	0



$\overline{AB} = \overline{A} + \overline{B}$

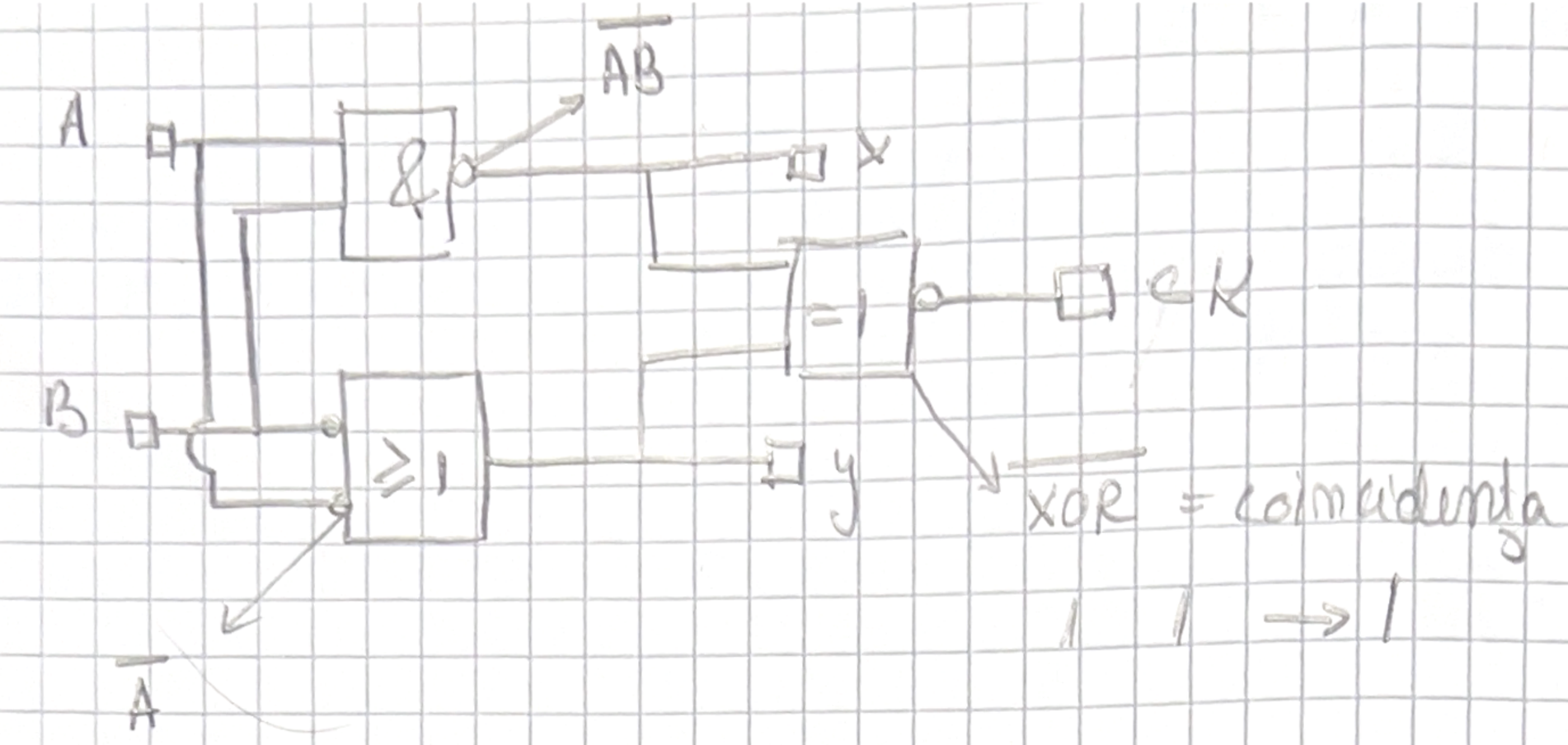
($\overline{AB}$ = x ; $\overline{A} + \overline{B}$ = y)

$x, y : \Sigma_2 \rightarrow \Sigma_2$

$x = \overline{AB}$

$y = \overline{A} + \overline{B}$

A	B	$\overline{A}$	$\overline{B}$	$\overline{AB}$	x	y
0	0	1	1	0	1	1
0	1	1	0	0	1	1
1	0	0	1	0	1	1
1	1	0	0	1	0	0



XOR

A	B	XOR $A \oplus B$	XNOR $\overline{A \oplus B}$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	0	1

LP3

Formele canonice

FCD - Forma canonică disjunctă

SoP - Sum of Products

mintermi: stările, scrise ca f_m și (and) între ele. de intrare la poziția pt. care funcția are val. de ieșire 1.

$m(x_1, x_2, \dots, x_n)$, x_n indicele poziției pt. care $y = 1$

$\Sigma (x_1, x_2 \dots x_n)$

Eg.

$y = m(1, 2, 4, 6)$ în poz 1, 2, 4, 6, punctia este 1

$y: \Sigma_2^3 \longrightarrow \Sigma_2$

#	A (MSB)	B	C (LSB)	y
0	0	0	0	0
1	0	0	1	1 (1)
2	0	1	0	1 (2)
3	0	1	1	0
4	1	0	0	1 (4)
5	1	0	1	0
6	1	1	0	1 (6)
7	1	1	1	0

$y = m(1,2,4,6) = \overline{A}\,\overline{B}C + A\overline{B}\,\overline{C} + A\overline{B}\overline{C} + AB\overline{C}$

(1) (2) (4) (6)

↗ Forma canonică disjunctivă [FCD]

Reducem forma canonică disjunctivă

$\overline{A}(\overline{B}C + B\overline{C}) + A\overline{C}(\overline{B} + B) = \overline{A}(B \oplus C) + A\overline{C}$

sau — xor — sau

Forma canonică Conjunctivă

Se scrie sub forma de produs de sume (PoS), pt stările în care fm ia valoarea 0 la fel ca pt termen. de intrare se iau negați

- Maxtermi

$M(x_1, x_2, \dots x_n)$

$\overline{M}(x_1, x_2 \dots x_n)$

Înșirată, nr. minitermilor și al maxtermilor nu vor da dimensiunea spațiului vectorial fm logică este definită (mapată)

Eg $y = FCD = m(1,2,4,6)$

$y = FCC = M(0,3,5,7)$

$y : \mathbb{Z}_2^3 \to \mathbb{Z}_2$

#	A	B	C	y
0	0	0	0	0 (0)
1	0	0	1	1
2	0	1	0	1
3	0	1	1	0 (3)
4	1	0	0	1
5	1	0	1	0 (5)
6	1	1	0	1
7	1	1	1	0 (7)

FCC: $(A+\underset{(2)}{B}+C)(A+\underset{(3)}{\bar{B}}+\bar{C})(\bar{A}+\underset{(5)}{B}+\bar{C})(\bar{A}+\bar{B}+\bar{C})$ [FCC]

Reducem FCC

$y: A + A\bar{B} + A\bar{C} + AB + B\bar{B} + B\bar{C} + AC + \bar{A}\bar{B} + C\bar{C}) \dots$

$= (A + B\bar{C} + C\bar{B})(\bar{A}+B+\bar{C})(\bar{A}+\bar{B}+\bar{C}) =$

$= (A\bar{A} + AB + A\bar{C} + AB\bar{C} + BB\bar{C} + B\bar{C}\bar{C} + \bar{A}C\bar{B} + C\bar{B}B + C\bar{B}\bar{C})$

$= (AB + A\bar{C} + AB\bar{C} + B\bar{C} + AC\bar{B})(\bar{A}+B+\bar{C})$

$A\bar{A} = 0$
$B\bar{B} = 0$
$C\bar{C} = 0$

$= A\bar{A}B + AB\bar{B} + AB\bar{C} + A\bar{A}\bar{C} + AB\bar{C} + A\bar{C} + \bar{A}B\bar{C} + AB\bar{B}C +$
$+ A\bar{B}\bar{C} + \bar{A}B\bar{C} + B\bar{C} + B\bar{B}\bar{C} + \bar{A}C\bar{B} + \bar{A}B\bar{C} + \bar{A}B\bar{C}\bar{C}$

$= AB\bar{C} + AB\bar{C} + A\bar{C} + AB\bar{C} + B\bar{C} + \bar{A}B\bar{C} =$

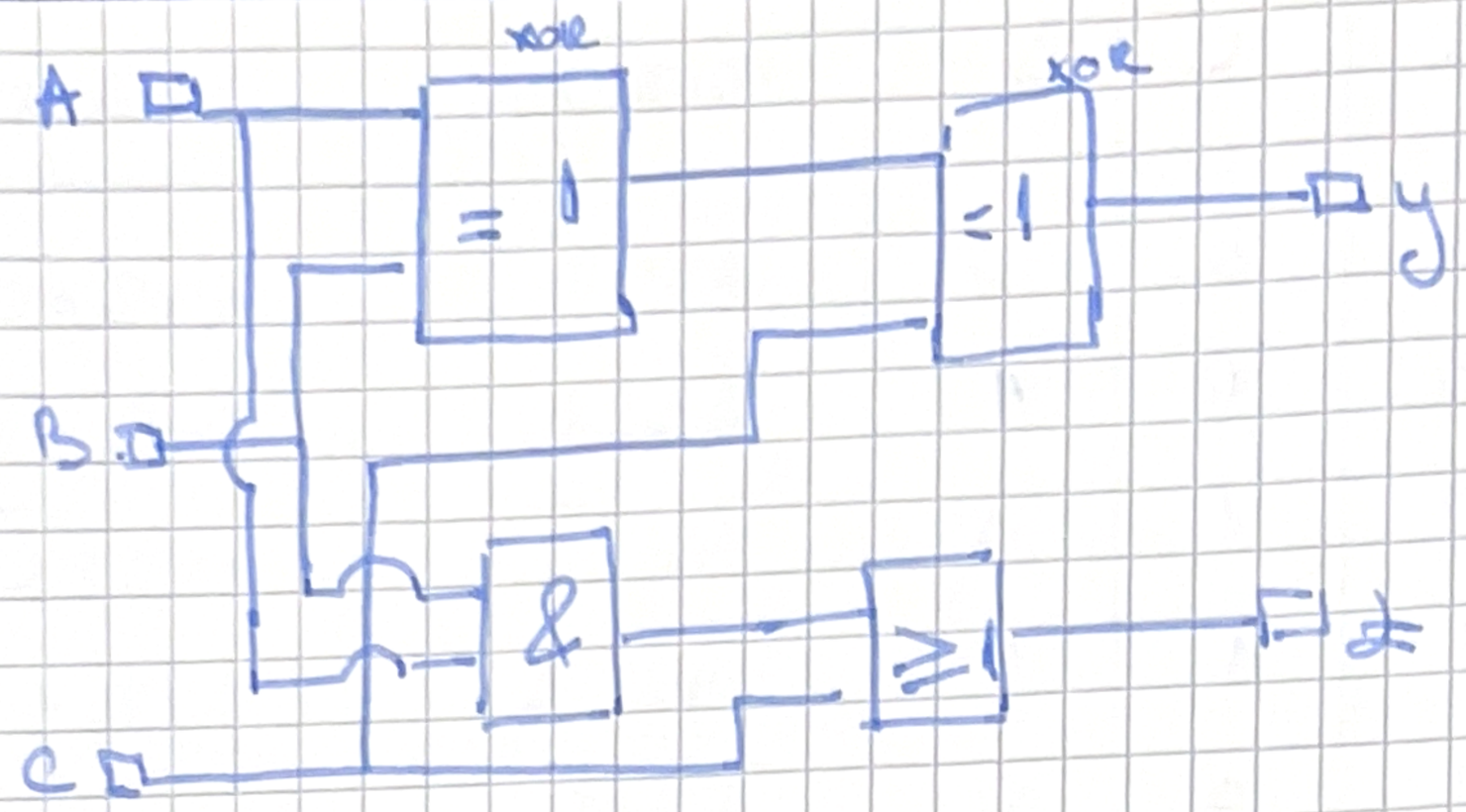
$= A(B\bar{C} + B\bar{C} + \bar{C}) + \bar{A}(\bar{B}\bar{C} + B\bar{C})$

#	A	B	C	D	minterm	maxterm
0	0	0	0	0	$\bar{A}\bar{B}\bar{C}\bar{D}$	$A+B+C+D$
1	0	0	0	1	$\bar{A}\bar{B}\bar{C}D$	$A+B+C+\bar{D}$
2	0	0	1	0	$\bar{A}\bar{B}C\bar{D}$	$A+B+\bar{C}+D$
3	0	0	1	1	$\bar{A}\bar{B}CD$	$A+B+\bar{C}+\bar{D}$
4	0	1	0	0	$\bar{A}B\bar{C}\bar{D}$	$A+\bar{B}+C+D$
5	0	1	0	1	$\bar{A}B\bar{C}D$	$A+\bar{B}+C+\bar{D}$
6	0	1	1	0	$\bar{A}BC\bar{D}$	$A+\bar{B}+\bar{C}+D$
7	0	1	1	1	$\bar{A}BCD$	$A+\bar{B}+\bar{C}+\bar{D}$
8	1	0	0	0	$A\bar{B}\bar{C}\bar{D}$	$\bar{A}+B+C+D$
9	1	0	0	1	$A\bar{B}\bar{C}D$	$\bar{A}+B+C+\bar{D}$
A (10)	1	0	1	0	$A\bar{B}C\bar{D}$	$\bar{A}+B+\bar{C}+D$
B (11)	1	0	1	1	$A\bar{B}CD$	$\bar{A}+B+\bar{C}+\bar{D}$
C (12)	1	1	0	0	$AB\bar{C}\bar{D}$	$\bar{A}+\bar{B}+C+D$
D (13)	1	1	0	1	$AB\bar{C}D$	$\bar{A}+\bar{B}+C+\bar{D}$
E (14)	1	1	1	0	$ABC\bar{D}$	$\bar{A}+\bar{B}+\bar{C}+D$
F (15)	1	1	1	1	$ABCD$	$\bar{A}+\bar{B}+\bar{C}+\bar{D}$

Pt. a defini o fn logică:

- forma algebrică
- tb. de adevăr
- formele canonice
- diagrama asociată

Reducem nr. de operații pt. a ajunge la forma optimizată.



$$\#, y : \Sigma_2^2 \to \Sigma_2$$

$$y : (A \oplus B) \oplus C = m(1,2,4,7) = M(0,3,5,6)$$

$$z = AB + C = m(1,3,5,6,7) = M(0,2,4)$$

A	B	C	A⊕B	y	AB	z
0	0	0	0	0	0	0
0	0	1	0	1	0	1
0	1	0	1	1	0	0
0	1	1	1	0	0	1
1	0	0	1	1	0	0
1	0	1	1	0	0	1
1	1	0	0	0	1	1
1	1	1	0	1	1	1